

ES-EL-013.8kV-001 : Medium-Voltage Metal-Enclosed Switchgear (13.8 kV)

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[SECTION 1] Scope

This Standard defines the minimum technical requirements for indoor, metal-clad, withdrawable vacuum circuit breaker switchgear intended for 13.8 kV, 60 Hz systems at mining and processing facilities in the Kingdom of Saudi Arabia.

The scope covers design, fabrication, type and routine testing, delivery, installation, commissioning, documentation, spare parts, and quality control. Interfaces include utility interconnection at 13.8 kV, plant feeders, protection and control systems, and earthing systems.

The Standard applies to complete switchgear lineups including incomers, bus sections, bus couplers, and feeder panels. Tailoring for specific projects is permitted where explicitly stated in project data sheets.

[SECTION 2] Normative References

IEC 62271-1, IEC 62271-200, IEC 62271-100, IEC 62271-102 (switchgear).

IEC 60529 (ingress protection). IEC 60068 (environmental testing). IEC 61850 (substation communication).

IEC 60044 and IEC 60076 for instrument transformers (CT, VT). ISO 12944 for corrosion protection. Saudi Building Code (SBC) for seismic. SEC and SASO compliance as applicable.

If a conflict is identified, statutory requirements take precedence. Otherwise this Standard governs.

[SECTION 3] Service Conditions

Ambient temperature shall be from minus 5 C to plus 55 C. Switchroom continuous design temperature shall be plus 50 C minimum. The equipment shall operate within nameplate ratings at these conditions or the supplier shall state derating explicitly.

Relative humidity up to 95 percent non condensing. Altitude less than or equal to 1000 m above sea level unless otherwise specified. Dust and sand exposure per IEC 60068 2 68.

For coastal and desert locations, corrosion category C4 or C5 per ISO 12944 may apply. The coating system and surface preparation shall be selected accordingly and recorded in the quality documentation.

[SECTION 4] Electrical Ratings

Rated voltage: 17.5 kV class suitable for 13.8 kV service. Frequency 60 Hz.

Insulation performance: lightning impulse LI BIL greater than or equal to 95 kV and power frequency AC withstand greater than or equal to 38 kV for 1 minute.

Short circuit rating: not less than 31.5 kA for 1 s. The making capacity shall comply with IEC 62271 requirements for the selected rating.

Main busbar continuous current not less than 2500 A. Temperature rise shall remain within IEC limits at the specified service conditions. Feeder ratings shall be as per the load list or the project data sheet.

Instrument transformers: protection CT accuracy class 5P20 or better; metering CT accuracy class 0.2S. VT accuracy class 0.5 or better for metering and control.

[SECTION 5] Internal Arc and Personnel Protection

The switchgear shall have an Internal Arc Classification IAC AFLR of at least 31.5 kA for 1 s for each functional unit. Type test reports from an accredited laboratory shall be provided.

Pressure relief shall be achieved via a top mounted exhaust and ducts routed to a safe discharge area. The general arrangement drawings shall show the routing concept, cross section, and termination location. The final routing shall be coordinated with the building and HVAC design.

Operator safety features shall include door closed racking, mechanical interlocks that prevent closing on an earthed circuit, and interlocks that prevent rack movement when the breaker is closed.

[SECTION 6] Ingress Protection and Compartmentation

External enclosure protection shall be IP4X minimum. Internal separation shall be IP2X between accessible compartments and live parts.

Construction shall be metal clad with separation not less than Form 3b or an equivalent arrangement providing at least the same degree of isolation and maintainability.

[SECTION 7] Primary Equipment

Vacuum circuit breakers shall be withdrawable. Visible isolation and an earthing switch with making capacity shall be provided.

Surge arresters shall be installed at incomers. Feeder equipment shall be provided per the project specification including current limiting fuses where required for auxiliary transformers.

[SECTION 8] Secondary Systems and Protection

Numerical relays shall support IEC 61850 Edition 2 station bus. Redundant Ethernet paths shall be provided using dual interfaces or redundant switches as specified.

Protection functions shall include overcurrent and earth fault, breaker failure protection, and bus protection where specified. Synch check shall be provided for sources and bus couplers as required.

Time synchronization shall use SNTP or PTP as required by the Substation Automation System. Disturbance records shall be exportable in COMTRADE format. Relay settings shall be provided in both native and exchange formats.

[SECTION 9] Earthing and Bonding

A copper earth bar of continuous construction shall run the full length of the lineup. Doors, CT and VT cases, and removable elements shall be bonded using flexible conductors.

The target substation earth resistance is 1 ohm or less unless otherwise stated. Connection to the building earth grid shall follow the construction drawings and site earthing philosophy.

[SECTION 10] Installation Environment and Seismic

Provide clearances, access envelopes, and lifting points on the general arrangement drawings. Ventilation shall be sized for a room temperature of plus 50 C with consideration for dust loading of filters.

Cable entry details, gland plates, and segregation shall be coordinated with the cable schedule. Access to the cable room and safe maintenance space in front and rear shall be provided.

Seismic bracing and anchorage shall comply with the Saudi Building Code category applicable to the site. Calculations and base details shall be submitted for review.

[SECTION 11] Quality, Coatings, and Testing

A project specific Quality Plan and an Inspection and Test Plan shall define hold and witness points for FAT and SAT. The plan shall reference the applicable clauses of this Standard and the manufacturer procedures.

Routine tests shall include high voltage withstand, insulation resistance, functional checks, and wiring verification. Results shall be included in the FAT report pack.

IEC 61850 functional testing shall be part of FAT where IEC 61850 is in scope.

Demonstrations shall include GOOSE publications and subscriptions, MMS connectivity, and time synchronization.

Type test evidence shall cover dielectric, short time current, and internal arc performance. Coating systems shall be qualified for ISO 12944 C4 or C5 as applicable and inspection records shall state surface preparation and dry film thickness.

[SECTION 12] Documentation Deliverables

General arrangement drawings, single line diagrams, and detailed wiring schematics.

Protection and coordination study with CT saturation checks at maximum fault levels. Relay settings files in native and exchange formats.

Arc flash assessment and label set conforming to the site template. CT and VT data sheets. FAT and SAT protocols and reports. Operation and maintenance manuals. Recommended spares list.

[SECTION 13] Deviations

Any deviation from this Standard shall be submitted on the Company Deviation Request Form. Acceptance is valid only when provided in writing by the Company.

[SECTION 14] Marking and Compliance

Nameplates shall be provided in English and Arabic. SEC and SASO compliance shall be demonstrated where applicable. Hazardous area classification is not applicable unless otherwise specified in the project data sheet.

Annex A : Clause Checklist (Normative)

The supplier shall submit a clause by clause compliance matrix with the following fields: Clause, Requirement, Vendor Claim, Evidence, Status, Comment, Proposed Action, Owner.

Where a requirement is not met in full, the matrix shall identify the deviation, the technical impact, and the proposed mitigation or alternative, together with references to evidence documents.